

W1 PRACTICE

PART 3 – MANIPULATION

⚠ Before this practice

- You need to have completed: **PART 1- EXPLORATION**

🧠 Learning objectives

- Apply type **inference** for variable declarations.
- Handle **nullable** and **non-nullable** variables.
- Differentiate between **final** and **const**.
- Manipulate **strings, lists, and maps**.
- Use **loops** and **conditions** to control flow.
- Define and call functions with positional and **named arguments**, understand **arrow syntax**



L0 NO AI

Solve problems entirely on your own.

📖 Are you lost?

Read the following documentation to be ready for this practice:

- | | | |
|----------------------------------|------------------------------|-----------------------------|
| ✓ Variables | ✓ Lists | ✓ Functions |
| ✓ Null Safety | ✓ Loops | |
| ✓ Built-in types | ✓ Conditions | |



EX 1 – Manipulate Types

Are you clear about strings, integer, list, map, set, objects in Dart?

Examine the given data structures and **write the inferred Dart type** for each one (see example)

Notes

- **First find by yourself the type**
- *If you need, double check your answer with VSCode.*

Data	Dart Type
<pre>const studentGrades = { 'Sokan': [90, 85, 88], 'Sokea': [70, 80, 75], 'Hay': [95, 92, 89], };</pre>	Map<String, List<int>>
<pre>const pizzaPrices = { 'margherita': 5.5, 'pepperoni': 7.5, 'vegetarian': 6.5, };</pre>	Map<String, double>
<pre>const books = [{'title': '1984', 'author': 'George Orwell'}, {'title': 'Brave New World', 'author': 'Aldous Huxley'}, {'title': 'Fahrenheit 451', 'author': 'Ray Bradbury'},];</pre>	List<map<String,String>>
<pre>const company = { 'HR': {'employees': 5, 'budget': 20000}, 'IT': {'employees': 10, 'budget': 50000}, 'Marketing': {'employees': 7, 'budget': 30000}, };</pre>	Map<String,map<String, int>>
<pre>const coordinates = [[1, 2], [3, 4], [5, 6],];</pre>	List<list<int>>
<pre>const productDetails = { 'id': 101, 'name': 'Laptop', 'price': 999.99, 'inStock': true, };</pre>	Map<String,object>
<pre>const operations = [(int a, int b) => a + b, (int a, int b) => a - b, (int a, int b) => a * b,];</pre>	List<int function(int,int)>
<pre>const distances = {3.1, 5.5, 10.2, 7.8};</pre>	Set<double>
<pre>const university = { 'departments': [{ 'name': 'Computer Science', 'students': [{'name': 'Alice', 'age': 22}, {'name': 'Bob', 'age': 24},] }, { 'name': 'Mathematics', 'students': [{'name': 'Charlie', 'age': 21}, {'name': 'Dave', 'age': 23},] }] };</pre>	Map<String, List<Map<String, Object>>>

EX 2 – Manipulate final and const

In this exercise, you need to decide which variable can be declared as **const** or **final**.

```
// 1 - startText
String iLike = 'I like pizza';

// 2 - toppings
String toppings = 'with tomatoes';
toppings += " and cheese";

// 3 - message
String message = '$iLike $toppings';
print(message);

// 4 - rgbColors
List<String> rgbColors = ['red', 'blue', 'green'];
print(rgbColors);

// 5 - weekDays
List<String> weekDays = ['monday', 'tuesday', 'wednesday'];
weekDays.add('thursday');
print(weekDays);

// 6 - scores
List<int> scores = [45,35,50];
scores = [45,35,50, 78];
print(scores);
```

Guess which variables can be declared as **const**, **final** or **var**, and explain your choices.

Notes

- Read [here](#) to understand the concepts.
- Prefer const over final over var.

	VAR, FINAL, CONST?	WHY
iLike	CONST	Because this variable never changes
toppings	var	Because we add another string to it
message	var	Because we combine 2 variable
rgbColors	const	Because rgb is only red green and blue
weekDays	final	Because we want to add another value
score	var	Because we want to reassign the list